Quantum Optimal Transport Amplitude Method (QOTAM)

Mathematical derivations appendix

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This appendix supplies the calculations cited as S.1–S.8 in the revised main article. Standard identities are attributed locally; their finite-regulator applications, assumptions, and numerical interpretations are stated explicitly. Section S.8 documents the interacting calculation and the comparison with consistently truncated first- and second-order coupling expansions. The accompanying editorial notes track changes from cjp-2026-0366.

S Mathematical derivations and assumptions

This appendix supplies intermediate steps for the regulated-measure, transport, correlator, and finite-dimensional operator equations used in the main text. The Gibbs identity, entropy disintegration, and scattering reduction are established mathematical tools; their use here does not make them new QOTAM theorems. Definitions of the QOTAM representation are distinguished below from consequences of those definitions. Unless otherwise stated, $\hbar > 0$ is retained explicitly and logarithms are natural. All measures appearing as priors in classical relative entropy are probability measures. Statements about continuous field variables are not inferred merely from the finiteness of the number of lattice sites.

S.1 Gaussian reference and normalized source-dependent measure

A finite-mode bosonic regulator supplies coordinates $\Phi \in \mathbb{R}^{d_N}$. Choosing a real symmetric positive-definite reference matrix A_N defines $S_{0,N}(\Phi) = \Phi^{\mathsf{T}} A_N \Phi/2$. This is a reference choice, not a uniqueness claim about the free/interacting split. Diagonalize $A_N = O^{\mathsf{T}} \operatorname{diag}(a_1, \dots, a_{d_N}) O$, where $a_j > 0$ and O is orthogonal. Changing variables to $u = O\Phi$ gives

$$\begin{aligned} Z_{0,N} &= \int_{\mathbb{R}^{d_N}} e^{-\Phi^{\mathsf{T}} A_N \Phi/(2\hbar)}\, \mathrm{d}\Phi = \prod_{j=1}^{d_N} \int_{\mathbb{R}} e^{-a_j u_j^2/(2\hbar)}\, \mathrm{d}u_j \\ &= \frac{(2\pi\hbar)^{d_N/2}}{\sqrt{\det A_N}}, \qquad \mathrm{d}R_{0,N} = Z_{0,N}^{-1} e^{-S_{0,N}/\hbar}\, \mathrm{d}\Phi. \end{aligned} \tag{S.1}$$

Completing the square in a real source J yields

$$\begin{aligned} -\tfrac{1}{2}\Phi^{\mathsf{T}} A_N \Phi + J^{\mathsf{T}}\Phi &= -\tfrac{1}{2}(\Phi - A_N^{-1}J)^{\mathsf{T}} A_N (\Phi - A_N^{-1}J) + \tfrac{1}{2} J^{\mathsf{T}} A_N^{-1} J, \\ \mathbb{E}_{R_{0,N}} e^{J\cdot\Phi/\hbar} &= e^{J^{\mathsf{T}} A_N^{-1} J/(2\hbar)}, \qquad \mathbb{E}_{R_{0,N}}[\Phi_i \Phi_j] = \hbar (A_N^{-1})_{ij}. \end{aligned} \tag{S.2}$$

Thus the covariance, including its factor of $\hbar$, follows by differentiation of a finite Gaussian integral.

Write the microscopic regulated action as $S_{E,N} = S_{0,N} + V_N$ and put $U_J = V_N - J\cdot\Phi$. The normalized interacting law is defined by

$$\mathcal{Z}_N[J] = \mathbb{E}_{R_{0,N}} e^{-U_J/\hbar}, \qquad \frac{\mathrm{d}P_{N,J}}{\mathrm{d}R_{0,N}} = \frac{e^{-U_J/\hbar}}{\mathcal{Z}_N[J]}. \tag{S.3}$$

We require U_J to be finite $R_{0,N}$-almost everywhere and $0 < \mathcal{Z}_N[J] < \infty$. For the finite-valued variational statements below we additionally require $\mathbb{E}_{P_{N,J}}|U_J| < \infty$. These conditions hold for the finite-dimensional stable polynomial scalar example with a positive quartic term and real sources in a bounded neighborhood of zero. They do not follow from finite dimension for an arbitrary interaction. The complete unnormalized partition integral is $Z_{0,N}\mathcal{Z}_N[J]$; the factor $Z_{0,N}$ is independent of J.

S.2 Gibbs identity, its minimizer, and reference-split invariance

Suppress N temporarily. On the domain of trial probabilities $Q \ll R_0$ for which $D_{\mathrm{KL}}(Q\|R_0) < \infty$ and $\mathbb{E}_Q|U_J| < \infty$, choose the free-energy representation

$$\mathfrak{F}[Q;J] = \mathbb{E}_Q U_J + \hbar D_{\mathrm{KL}}(Q\|R_0). \tag{S.4}$$

This is the Gibbs relative-entropy representation, not the only possible objective with the same minimizer. The entropy conventions and path-space setting are consistent with Ref. [1]; stochastic control representations such as Ref. [2] require their own additional assumptions.

By Eq. (S.3),

$$\begin{aligned} \log\frac{\mathrm{d}Q}{\mathrm{d}P_J} &= \log\frac{\mathrm{d}Q}{\mathrm{d}R_0} + \frac{U_J}{\hbar} + \log\mathcal{Z}[J], \\ \hbar D_{\mathrm{KL}}(Q\|P_J) &= \hbar D_{\mathrm{KL}}(Q\|R_0) + \mathbb{E}_Q U_J + \hbar\log\mathcal{Z}[J]. \end{aligned} \tag{S.5}$$

Consequently

$$\mathfrak{F}[Q;J] = -\hbar\log\mathcal{Z}[J] + \hbar D_{\mathrm{KL}}(Q\|P_J), \qquad \inf_Q \mathfrak{F}[Q;J] = -\hbar\log\mathcal{Z}[J]. \tag{S.6}$$

The integrability conditions above ensure that P_J itself is admissible. Nonnegativity of relative entropy then proves attainment and uniqueness: $Q_\star = P_J$. An arbitrary density is not claimed to have finite entropy.

For an interior density $q = \mathrm{d}Q/\mathrm{d}R_0$, one can also vary the normalization constraint. The Lagrangian and its first variation are

$$\begin{aligned} \mathcal{L}[q,\zeta] &= \int (U_J q + \hbar q\log q)\,\mathrm{d}R_0 + \zeta\left(\int q\,\mathrm{d}R_0 - 1\right), \\ \delta\mathcal{L} &= \int [U_J + \hbar(\log q + 1) + \zeta]\,\delta q\,\mathrm{d}R_0. \end{aligned} \tag{S.7}$$

Stationarity gives $q \propto e^{-U_J/\hbar}$, and normalization fixes the constant. The nonnegativity proof, rather than stationarity alone, establishes the global minimum.

If two normalized references R_A, R_B represent the same P_J, the corresponding functionals satisfy, on their common admissible domain,

$$\mathfrak{F}_A[Q;J] - \mathfrak{F}_B[Q;J] = \hbar\log\frac{\mathcal{Z}_B[J]}{\mathcal{Z}_A[J]}, \tag{S.8}$$

which is independent of Q. When the two reference actions are source independent and their sum with their respective interactions is the same microscopic action, $Z_{0,A}\mathcal{Z}_A[J] = Z_{0,B}\mathcal{Z}_B[J]$. The difference in Eq. (S.8) is then also independent of J, so source derivatives agree. Equality of optimizers at each J alone would not justify dropping a source-dependent normalization from a generating functional.

S.3 Boundary constraints, disintegration, cost, and duality

Let $T = (B_-, B_+)$ be a measurable map from field histories to two boundary spaces. Assume standard Borel spaces so that regular conditional probabilities exist. For $P = P_{N,0}$ and $Q \ll P$, define $R_S = T_\# P$ and $\gamma = T_\# Q$. Disintegrations read

$$P(\mathrm{d}\Phi) = P^{xy}(\mathrm{d}\Phi)R_S(\mathrm{d}x\mathrm{d}y), \qquad Q(\mathrm{d}\Phi) = Q^{xy}(\mathrm{d}\Phi)\gamma(\mathrm{d}x\mathrm{d}y). \tag{S.9}$$

Write $r = \mathrm{d}\gamma/\mathrm{d}R_S$ and $h_{xy} = \mathrm{d}Q^{xy}/\mathrm{d}P^{xy}$. Integrating a test function against $r(T\Phi)h_{T\Phi}(\Phi)P(\mathrm{d}\Phi)$ reproduces its Q integral. Hence

$$\frac{\mathrm{d}Q}{\mathrm{d}P}(\Phi) = r(T\Phi)h_{T\Phi}(\Phi) \quad P\text{-almost everywhere.} \tag{S.10}$$

Taking a logarithm and integrating gives the entropy chain rule

$$D_{\mathrm{KL}}(Q\|P) = D_{\mathrm{KL}}(\gamma\|R_S) + \int D_{\mathrm{KL}}(Q^{xy}\|P^{xy})\,\gamma(\mathrm{d}x\mathrm{d}y). \tag{S.11}$$

This is the established path-to-endpoint reduction of the Schrödinger problem; see Ref. [1], Sec. 2.

For fixed γ, the second term is minimized by $Q^{xy} = P^{xy}$. The constrained path minimum therefore reduces to

$$\inf_{\substack{Q:(B_-)_\# Q=\rho_- \\ (B_+)_\# Q=\rho_+}} D_{\mathrm{KL}}(Q\|P) = \inf_{\gamma\in\Pi(\rho_-,\rho_+)} D_{\mathrm{KL}}(\gamma\|R_S). \tag{S.12}$$

This equality does not assert existence without assumptions. Existence and strict positivity are immediate in the strictly positive finite discrete setting proved below; continuous existence requires the appropriate compactness and entropy conditions.

The constrained optimizer generally differs from P: it equals P precisely when the prescribed marginals equal those of P, assuming the preceding minimizers exist. If $\mathrm{d}\gamma_\star = f(x)g(y)\mathrm{d}R_S$ with $f, g > 0$, then

$$\frac{\mathrm{d}Q_\star}{\mathrm{d}P}(\Phi) = f(B_-\Phi)g(B_+\Phi), \qquad \mathbb{E}_P O = \mathbb{E}_{Q_\star}\left[\frac{O}{f(B_-)g(B_+)}\right] \tag{S.13}$$

for integrable observables. Thus obtaining the original theory's expectation from a changed bridge requires the appropriate reweighting; variance reduction is not implied by this identity.

Cost and reference-measure changes. Take a product probability measure $\mu = m_- \otimes m_+$ with $r_S = \mathrm{d}R_S/\mathrm{d}\mu > 0$ and $q = \mathrm{d}\gamma/\mathrm{d}\mu$. Direct substitution gives

$$\begin{aligned}\hbar D_{\mathrm{KL}}(\gamma\|R_S) &= \hbar \int q\log(q/r_S)\,\mathrm{d}\mu \\ &= \int c_S\,\mathrm{d}\gamma + \hbar D_{\mathrm{KL}}(\gamma\|\mu), \qquad c_S = -\hbar\log r_S.\end{aligned} \tag{S.14}$$

For normalized changed references $m'_- = Z_a^{-1}e^{-a/\hbar}m_-$ and $m'_+ = Z_b^{-1}e^{-b/\hbar}m_+$, holding R_S fixed gives

$$\begin{aligned}r'_S &= Z_a Z_b e^{(a+b)/\hbar} r_S, \\ c'_S &= c_S - a - b - \hbar\log(Z_a Z_b).\end{aligned} \tag{S.15}$$

The entropy term in Eq. (S.14) changes by the opposite amount, so the full objective is unchanged. The opposite change of the entropy term below verifies the sign directly. Separately, adding $a(x) + b(y)$ to a cost while keeping μ fixed shifts a fixed-marginal objective by the constant $\int a\,\mathrm{d}\rho_- + \int b\,\mathrm{d}\rho_+$. These are distinct bookkeeping operations. The choice to measure this free energy in action units produces the coefficient $\hbar$; dividing the complete objective by $\hbar$ leaves its optimizer unchanged.

Euler equation and the constant in the dual. Assume $\rho_- = r_- m_-$ and $\rho_+ = r_+ m_+$. With multipliers α, β, the Lagrangian is

$$\mathcal{L}(q,\alpha,\beta) = \int [c_S q + \hbar q\log q - (\alpha+\beta)q]\mathrm{d}\mu + \int \alpha\,\mathrm{d}\rho_- + \int \beta\,\mathrm{d}\rho_+. \tag{S.16}$$

At an interior optimum,

$$c_S + \hbar(\log q + 1) - \alpha - \beta = 0, \qquad q_\star = \exp\left[\frac{\alpha + \beta - c_S}{\hbar} - 1\right]. \tag{S.17}$$

The pointwise infimum over $q \geq 0$ yields the dual function

$$\mathcal{D}(\alpha, \beta) = \int \alpha \, \mathrm{d}\rho_- + \int \beta \, \mathrm{d}\rho_+ - \hbar \int \exp\left[\frac{\alpha + \beta - c_S}{\hbar} - 1\right] \mathrm{d}\mu. \tag{S.18}$$

Keeping the -1 makes the dual consistent with the primal entropy $q \log q$. Using $q(\log q - 1)$ instead changes the feasible primal objective by $-\hbar$ and removes this -1 from the exponential. Alternatively the multipliers can be shifted explicitly. A strong-duality claim needs its usual feasibility conditions.

Finite Schrödinger system. For $R_{ij} > 0$, $r_i, s_j > 0$, and equal normalized marginal masses, the transport polytope is compact and nonempty. The objective $\sum_{ij} \gamma_{ij} \log(\gamma_{ij}/R_{ij})$ is continuous, with $0 \log 0 = 0$, and strictly convex, so its minimizer exists and is unique. It has no zero entry: mixing any boundary candidate with the strictly positive feasible plan $r_i s_j$ produces a negative $t \log t$ contribution at a zero entry which dominates the finite first-order changes at positive entries. Interior stationarity then gives

$$\gamma_{ij} = a_i R_{ij} b_j, \qquad a_i = \frac{r_i}{\sum_j R_{ij} b_j}, \qquad b_j = \frac{s_j}{\sum_i a_i R_{ij}}. \tag{S.19}$$

Only $(a, b) \mapsto (ta, t^{-1}b)$ leaves the factors undetermined when the full positive bipartite support is connected. The original cost and marginals, rather than the factors individually, determine the coupling. This is the setting in which standard Sinkhorn convergence results apply[3].

S.4 Sources, cumulants, and fixed-endpoint kernels

Assume a neighborhood of real sources admitting a common integrable domination for the derivatives below. Finite dimension by itself does not justify interchanging derivatives and integration. From Eq. (S.3),

$$\partial_{J_i} \log \mathcal{Z}[J] = \frac{\mathbb{E}_{P_J} \Phi_i}{\hbar}, \qquad \partial_{J_i} \log \frac{\mathrm{d}P_J}{\mathrm{d}R_0} = \frac{\Phi_i - \mathbb{E}_{P_J} \Phi_i}{\hbar}. \tag{S.20}$$

For an observable O without explicit source dependence this gives

$$\partial_{J_i} \mathbb{E}_{P_J} O = \frac{\mathrm{Cov}_{P_J}(O, \Phi_i)}{\hbar}. \tag{S.21}$$

Set $\mathfrak{V}[J] = \inf_Q \mathfrak{F}[Q; J]$ and $W_E[J] = -\mathfrak{V}[J] = \hbar \log \mathcal{Z}[J]$. Repeated differentiation yields

$$\begin{aligned} -\partial_{J_i} \mathfrak{V}[J] &= \mathbb{E}_{P_J} \Phi_i, \\ -\hbar \partial_{J_i} \partial_{J_j} \mathfrak{V}[J] &= \mathbb{E}_{P_J}(\Phi_i \Phi_j) - \mathbb{E}_{P_J} \Phi_i \, \mathbb{E}_{P_J} \Phi_j, \\ G^{(n)}_{c,E;i_1 \ldots i_n} &= \hbar^{n-1} \partial_{J_{i_1}} \cdots \partial_{J_{i_n}} W_E[J] = -\hbar^{n-1} \partial_{J_{i_1}} \cdots \partial_{J_{i_n}} \mathfrak{V}[J]. \end{aligned} \tag{S.22}$$

These identities differentiate the *unconstrained* value. They do not replace it by the value of an arbitrary fixed-marginal bridge.

To generate perturbative coefficients introduce ξ multiplying V and use cumulants of the Gaussian reference:

$$-\hbar \log \mathbb{E}_{R_0} e^{-\xi V/\hbar} = \sum_{k \geq 1} \frac{(-1)^{k+1} \xi^k}{k! \hbar^{k-1}} \kappa_{k,0}(V). \tag{S.23}$$

This denotes the perturbative coefficient expansion, not a general convergent series at nonzero coupling. Stable scalar polynomial theories can have finite moments while this coupling series is only asymptotic.

For a positive Euclidean kernel with well-defined conditioned reference bridges, write R_0^{xy} for the normalized free path law between x and y. The ratio of the two fixed-boundary integrals is

$$\frac{K_E^S(y,x;T)}{K_{0,E}(y,x;T)} = \mathbb{E}_{R_0^{xy}} e^{-V/\hbar}. \tag{S.24}$$

Applying the Gibbs identity to this conditional measure, with the same integrability assumptions, gives

$$\mathfrak{V}_S(x,y;T) = \inf_{Q^{xy}} \{\mathbb{E}_{Q^{xy}} V + \hbar D_{\mathrm{KL}}(Q^{xy} \| R_0^{xy})\}, \qquad K_E^S = K_{0,E} e^{-\mathfrak{V}_S/\hbar}. \tag{S.25}$$

For continuous endpoints these are statements about regular conditional kernels on their admissible support; one does not impose a point mass on the original unconditioned path law and assume its entropy is finite. Also R_S is an endpoint *law*, whereas K_E^S is a propagation kernel. Identifying their densities requires the boundary factors and normalization, for example $r_S(x,y) = C^{-1} u(x) K_E^S(y,x;T) v(y)$ when such a representation is supplied.

S.5 Gaussian kernel, scalar vertices, and scattering conventions

For the oscillator reference (oscillator mass set to one), the stationary path with $x(0) = x$ and $x(T) = y$ solves $\ddot{x}_{\mathrm{cl}} = \omega^2 x_{\mathrm{cl}}$:

$$x_{\mathrm{cl}}(\tau) = \frac{x \sinh[\omega(T-\tau)] + y \sinh(\omega\tau)}{\sinh(\omega T)}. \tag{S.26}$$

Integration by parts gives

$$S_{0,E}[x_{\mathrm{cl}}] = \tfrac{1}{2}[x_{\mathrm{cl}} \dot{x}_{\mathrm{cl}}]_0^T = \frac{\omega}{2\sinh(\omega T)}[(x^2+y^2)\cosh(\omega T) - 2xy]. \tag{S.27}$$

Writing $x(\tau) = x_{\mathrm{cl}}(\tau) + \eta(\tau)$ with $\eta(0) = \eta(T) = 0$ eliminates the cross term. The fluctuation eigenvalues are $(n\pi/T)^2 + \omega^2$. Relative to the free-particle determinant, the product $\prod_{n\geq 1}[1 + (\omega T/(n\pi))^2] = \sinh(\omega T)/(\omega T)$ gives

$$K_{0,E}(y,x;T) = \sqrt{\frac{\omega}{2\pi\hbar \sinh(\omega T)}}\, e^{-S_{0,E}[x_{\mathrm{cl}}]/\hbar}. \tag{S.28}$$

The free-particle normalization follows from the short-time delta-function limit. This establishes the oscillator cost used for the Gaussian illustration; it is not an interacting field-theory calculation.

Scalar contractions and factors of $\hbar$. In the following scattering formulas set $\hbar = c = 1$ and use the Minkowski metric $(+,-,-,-)$. Keeping $\hbar$ elsewhere is essential: with $C_0 = \hbar A^{-1}$, a quartic insertion contributes $-\lambda \int \prod_{j=1}^4 C_0(x_j - z)\, \mathrm{d}z/\hbar$, and a two-cubic exchange contributes $g^2 \int \prod_{e=1}^5 C_0(e)/\hbar^2$. The formulas without these vertex factors therefore use $\hbar = 1$.

The scalar action and vacuum about which it is expanded are specified first; a positive quartic coupling ensures finite-regulator stability, and tadpole or vacuum shifts must be treated consistently beyond the displayed tree calculation. Fourier transformation of the quadratic action gives

$$S_{0,M} = \frac{1}{2}\int \frac{\mathrm{d}^4 p}{(2\pi)^4} \phi(-p)(p^2 - m^2)\phi(p), \qquad G_{0,F}(p) = \frac{i}{p^2 - m^2 + i0}. \tag{S.29}$$

The differential operator follows from the action. The $+i0$ is the chosen vacuum/time-ordering boundary prescription, not a consequence of differentiating the real unregulated action alone.

For $S_{\mathrm{int,M}} = -\int (g\phi^3/3! + \lambda\phi^4/4!)\mathrm{d}^4x$, local functional differentiation cancels the factorials and gives vertices $-ig$ and $-i\lambda$. Wick contractions can be obtained directly by differentiating the Gaussian generating functional[4]. At first quartic order, the 4! assignments of external legs cancel 1/4!. For a fixed two–two partition at order g^2, assigning the two labeled external legs to each cubic vertex gives $(3{\cdot}2)^2 = 36$

contractions, and interchange of the two vertices gives another factor of two. Thus $72/[2!(3!)^2] = 1$ for each channel. The connected terms are

$$G^{(4)}_{c,E}|_{\lambda} = -\lambda \int \mathrm{d}^4 z \prod_{j=1}^{4} G_{0,E}(x_j - z),$$
$$G^{(4)}_{c,E}|_{g^2,s} = g^2 \int \mathrm{d}^4 z\, \mathrm{d}^4 w\, G_{0,E}(x_1 - z) G_{0,E}(x_2 - z) G_{0,E}(z - w) G_{0,E}(w - x_3) G_{0,E}(w - x_4), \tag{S.30}$$

plus the other two external partitions.

Define $S = I + iT$ and $\langle q|T|p\rangle = (2\pi)^4 \delta^{(4)}(P_q - P_p)\mathcal{M}(q,p)$. Continuation and external-leg amputation[5, 6] give

$$i\mathcal{M}_4 = -i\lambda + \sum_{x=s,t,u} (-ig)\frac{i}{x - m^2 + i0}(-ig), \qquad \mathcal{M}_4 = -\lambda - g^2 \sum_{x=s,t,u} \frac{1}{x - m^2 + i0}. \tag{S.31}$$

Near an isolated simple exchange pole, the residue of $i\mathcal{M}$ is the product of the two amputated vertices and the propagator numerator. For equal external masses m, the pole $s = m^2$ lies below the physical $s \geq 4m^2$ two-particle region; the residue statement refers to the analytic amplitude, not an accessible pole in that particular physical scan. Loop number is separately $L = I - V + 1$ for a connected graph; two cubic vertices connected by one internal line have $L = 0$.

For packet states the transition part of the S matrix in this convention is

$$\mathcal{A}_\Psi = \int \mathrm{d}\Pi_p\, \mathrm{d}\Pi_q\, \Psi^*_{\mathrm{out}}(q)(2\pi)^4 \delta^{(4)}(P_q - P_p) i\mathcal{M}(q,p)\Psi_{\mathrm{in}}(p). \tag{S.32}$$

For a constant contact kernel $\mathcal{A}_\Psi = -i\lambda\mathcal{I}_\Psi$; $\mathcal{M}_{4,\mathrm{ext}} = \mathcal{A}_\Psi/(i\mathcal{I}_\Psi)$ when $\mathcal{I}_\Psi \neq 0$. Neither packet normalization nor marginal normalization sets $\mathcal{I}_\Psi = 1$. The delta distribution is paired with packet functions before forming an absolute square.

S.6 Fermionic Gaussian algebra and the operator entropy projection

For d_F Grassmann pairs and an invertible matrix D, choose the Berezin order so that $\int \mathcal{D}\bar\psi\mathcal{D}\psi\, e^{-\bar\psi D\psi/\hbar} = \det(D/\hbar)$. Translating the Grassmann integration variables gives

$$\begin{aligned} &-\bar\psi D\psi + \bar\eta\psi + \bar\psi\eta \\ &\quad = -(\bar\psi - \bar\eta D^{-1})D(\psi - D^{-1}\eta) + \bar\eta D^{-1}\eta, \\ &\int \mathcal{D}\bar\psi\mathcal{D}\psi\, e^{(-\bar\psi D\psi + \bar\eta\psi + \bar\psi\eta)/\hbar} = \det(D/\hbar) e^{\bar\eta D^{-1}\eta/\hbar}. \end{aligned} \tag{S.33}$$

The ordered two-point contraction is $\hbar D^{-1}$. This algebraic Gaussian identity does not assert positivity of the determinant. In natural units the Dirac propagator is the boundary-value distribution

$$S_F(p) = \frac{i(\not{p} + m_\psi)}{p^2 - m_\psi^2 + i0}, \qquad (\not{p} - m_\psi)(\not{p} + m_\psi) = (p^2 - m_\psi^2)I. \tag{S.34}$$

The Clifford identity supplies the numerator; the causal prescription is specified on the scalar denominator. A literal finite regulator in $i/(\not{p} - m_\psi + i\eta)$ need not equal the expression obtained by inserting the same $i\eta$ into the scalar denominator.

The Yukawa interaction gives $-iy$. Ordered Grassmann contractions and fermionic external-leg amputation give the two terms

$$i\mathcal{M}_{\psi\phi\to\psi\phi} = (-iy)^2 \bar u(p')[S_F(p + k) + S_F(p - k')]u(p). \tag{S.35}$$

Exchanging two identical *incoming* fermions or two identical *outgoing* fermions reverses the ordered external state by CAR, and therefore reverses its matrix element. An interchange between incoming and outgoing legs is a crossing operation and is not described solely by this antisymmetry statement.

What reflection positivity supplies. For the chosen Euclidean regulator one must establish the unnormalized inequality $\mathcal{I}(e^{-S_E/\hbar}\Theta(F)F) \geq 0$ and a nonzero partition function before using normalized expectations. The free-overlap/chiral-Yukawa reflection-positivity result in Ref. [7] is an input theorem about the specified regulator, not a consequence of determinant positivity. It gives the positive semidefinite OS form and quotient construction

$$(F,G)_{\mathrm{OS}} = \langle \Theta(F)G\rangle_E, \qquad \mathcal{H}_{\mathrm{OS}} = \overline{\mathcal{A}_+/\mathcal{N}}, \qquad \mathcal{N} = \{F : (F,F)_{\mathrm{OS}} = 0\}. \tag{S.36}$$

A compatible time-translation reconstruction is an additional step. At nonzero lattice spacing time translations are initially discrete. If that reconstruction supplies a nonnegative self-adjoint contraction T_a, functional calculus defines $H_a = -(\hbar/a_\tau)\log T_a$ on its positive spectral support; zero spectral modes and any limit in time extent require explicit treatment. A continuous-time semigroup or an asymptotic particle Fock representation does not follow merely by writing a finite periodic lattice expectation. The relevant reconstruction hypotheses must be stated[8, 9]. This appendix does not add a new general reconstruction theorem.

Finite-cutoff Gibbs state and variation. Assume a parity-even, lower-bounded self-adjoint reconstructed Hamiltonian has been supplied, shift its lower bound to zero, and retain a nonzero finite-rank spectral subspace $P_{E_c} = \mathbf{1}_{[0,E_c]}(H_a)$ which respects fermion parity. A finite number of bosonic lattice variables does not by itself make this state space finite dimensional. On the retained space define

$$\Omega = \frac{e^{-\beta H_{a,E_c}/\hbar}}{Z}, \qquad Z = \operatorname{Tr} e^{-\beta H_{a,E_c}/\hbar}, \qquad \beta > 0. \tag{S.37}$$

Here β has units of Euclidean time. Diagonalizing H_{a,E_c} shows that all eigenvalues of Ω are strictly positive. For density matrices Γ,

$$\hbar S(\Gamma\|\Omega) = \beta \operatorname{Tr}(\Gamma H_{a,E_c}) + \hbar \operatorname{Tr}(\Gamma\log\Gamma) + \hbar\log Z. \tag{S.38}$$

Specify trace-preserving completely positive maps $\mathcal{B}_\pm$ to chosen boundary matrix algebras, and impose $\mathcal{B}_\pm(\Gamma) = \rho_\pm$ and parity invariance. The resulting feasible set, when nonempty, is compact and convex. Since $x\log x$ extends continuously to zero, the objective is continuous. Strict convexity of $\operatorname{Tr}(\Gamma\log\Gamma)$ and linearity of the other term prove existence and uniqueness. This is a finite matrix information-projection result in the setting of noncommutative entropic transport[10].

At an interior minimizer, for Hermitian variations X,

$$\delta\,\hbar S(\Gamma\|\Omega)[X] = \hbar \operatorname{Tr} X(\log\Gamma + I - \log\Omega). \tag{S.39}$$

With Hermitian multipliers $U_\pm$ for the boundary constraints, and with parity-compatible adjoint maps, stationarity yields

$$\Gamma_\star = \frac{\exp[\log\Omega + (\mathcal{B}_-^* U_- + \mathcal{B}_+^* U_+)/\hbar]}{\operatorname{Tr}\exp[\log\Omega + (\mathcal{B}_-^* U_- + \mathcal{B}_+^* U_+)/\hbar]}. \tag{S.40}$$

This interior expression is not asserted when the constraints force a singular optimizer. The exponential generally cannot be replaced by a product of noncommuting factors. Finally, constructing a Gibbs density matrix does not by itself construct a two-time process operator or prove that a constrained density projection generates the scattering amplitudes. Those identifications need a separate, specified dynamical construction.

S.7 Numerical channel equations and packet integration

For the channel calculation, set $D(x) = x - m^2 + i\eta$, with $\eta > 0$. At collocation nodes x_k with weights $w_k > 0$, write $K_N(x) = \sum_{n=0}^{N-1} a_n T_n(z(x))$. Define

$$B_{kn} = D(x_k)T_n(z(x_k)), \qquad b_k = -ig^2, \qquad W = \operatorname{diag}(w_1,\ldots,w_M). \tag{S.41}$$

The channel objective is $\|W^{1/2}(B\mathbf{a}-\mathbf{b})\|_2^2$. Taking the real differential, or the Wirtinger derivative with respect to $\mathbf{a}^*$, gives

$$B^\dagger W(B\mathbf{a}-\mathbf{b})=0. \tag{S.42}$$

Distinct nodes, $M \geq N$, and $D(x_k) \neq 0$ give full column rank of B for the polynomial basis. The unique least-squares solution should be computed by a weighted QR or singular-value decomposition, not by explicitly inverting the normal matrix. The inverse-propagator factor D is supplied; values of the closed solution need not be supplied.

If $R_N(x) = D(x)K_N(x) + ig^2$ and $K_\star = -ig^2/D$, the exact error identity is

$$K_N - K_\star = \frac{R_N}{D}, \qquad \|K_N - K_\star\|_{L^2(w)} \leq d_{\min}^{-1}\|R_N\|_{L^2(w)}, \quad d_{\min} = \inf_{x\in I}|D(x)| > 0. \tag{S.43}$$

For a normalized positive validation weight, Cauchy–Schwarz also bounds an average error by its root-mean-square error. Relative errors additionally need a nonzero reference denominator; cancellations can make that denominator small. These bounds certify a channel-response solve. They are not a derivation of that least-squares objective from the full Gibbs or bridge objective, and do not establish an OT-specific computational advantage.

Momentum measure and energy resolution. For labeled particles of equal mass, $\mathrm{d}\Pi(p) = \mathrm{d}^3\mathbf{p}/[(2\pi)^3 2E_\mathbf{p}]$. Conditioning the pair on zero total spatial momentum gives

$$\int \mathrm{d}\Pi(p_1)\mathrm{d}\Pi(p_2)(2\pi)^3\delta^{(3)}(\mathbf{p}_1+\mathbf{p}_2)F = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3 4E_\mathbf{p}^2}F(\mathbf{p},-\mathbf{p}). \tag{S.44}$$

This is a reduced center-of-mass measure; a normalizable complete packet state also requires its total-momentum dependence and the appropriate identical-particle symmetrization. On exact energy-conserving support $E_\mathbf{q} = E_\mathbf{p}$, one has $|\mathbf{q}| = |\mathbf{p}|$ and $s+t+u = 4m^2$.

Replacing $\delta(\Delta E)$ by $e^{-\Delta E^2/(2\sigma_E^2)}/(\sqrt{2\pi}\sigma_E)$ is an additional numerical regularization. Finite packet width by itself does not relax energy conservation of the infinite-time S matrix. A finite-width Gaussian can be interpreted using an explicitly specified finite-time switching prescription, or as a regulated integral whose $\sigma_E \to 0$ limit must be checked. For the finite-resolution definitions used in the retained numerical code, away from energy-conserving support,

$$s+t+u = 4m^2 + 4E_\mathbf{p}(E_\mathbf{p} - E_\mathbf{q}). \tag{S.45}$$

The finite-resolution integral is therefore not automatically an exactly energy-conserving scattering amplitude. At fixed σ_E, agreement between nested sample counts measures only a convergence difference, not an independently certified total error. Analytic reconstruction, regulator removal, and the relationship of these channel tests to a full entropy optimization remain separate requirements.

S.8 Direct interacting entropy minimization and independent quadrature

This subsection documents a numerical calculation added in the present revision; it is not a reinterpretation of the earlier scalar channel solve. Set $\hbar = 1$ and use three field coordinates $\phi = (x, y, z) \in [-L, L]^3$ with

$$S_0 = \tfrac{1}{2}(x^2+y^2+z^2) + \tfrac{\kappa}{2}[(y-x)^2 + (z-y)^2], \qquad V_\lambda = \frac{\lambda}{24}(x^4+y^4+z^4). \tag{S.46}$$

We take $\kappa = 0.7$, $L = 6$, and $\lambda = 0.5, 2, 4$. All interactions are retained. The domain cutoff is part of this finite numerical problem. It is not a four-dimensional QFT volume regulator or a claim of continuum scattering.

For positive tensor Gauss–Legendre weights $W_\nu = w_i w_j w_k$, define $r_{0,\nu} = W_\nu e^{-S_{0,\nu}}/Z_{0,h}$, where $Z_{0,h} = \sum_\nu W_\nu e^{-S_{0,\nu}}$. A trial vector of probability masses q minimizes

$$F_h(q) = \sum_\nu q_\nu[V_\nu + \log(q_\nu/r_{0,\nu})], \qquad q_\nu \geq 0, \quad \sum_\nu q_\nu = 1. \tag{S.47}$$

Its gradient on the positive simplex is $g_\nu = V_\nu + \log(q_\nu/r_{0,\nu}) + 1$. Entropic mirror descent with dimensionless step τ minimizes $\tau\langle g, q'\rangle + D_{\mathrm{KL}}(q'\|q)$ over q'. A normalization multiplier yields

$$q'_\nu = \frac{q_\nu e^{-\tau g_\nu}}{\sum_\mu q_\mu e^{-\tau g_\mu}}. \tag{S.48}$$

The implementation uses this gradient in log space, with $\tau = 1/4$ and 100 iterations. In this unrestricted finite simplex the update has a useful analytical check: starting at r_0, one finds inductively $q^{(k)} \propto r_0 e^{-a_k V}$ where $a_k = 1 - (1-\tau)^k$. Thus this test verifies the objective and its numerical implementation; it is not evidence for a new solver or favorable complexity. The scalar gradient RMS in Supplementary Fig. S1 is $[\sum q_\nu(g_\nu - \sum q_\mu g_\mu)^2]^{1/2}$.

After the solve, the exact discrete target may be evaluated independently as $p_{h,\nu} = r_{0,\nu}e^{-V_\nu}/\mathcal{Z}_h$. Equation (S.6) then checks $F_h(q) + \log \mathcal{Z}_h = D_{\mathrm{KL}}(q\|p_h)$. A fixed-grid optimization check does not measure field-quadrature error. We therefore repeat $n = 16, 24, 32, 48, 64, 80$ nodes per coordinate and compare against an adaptive integration which receives only Eq. (S.46).

Write $u_\lambda(x) = x^2/2 + \lambda x^4/24$ and $b_j(y) = \int_{-L}^{L} x^j e^{-u_\lambda(x) - \kappa(x-y)^2/2}\mathrm{d}x$. Conditional integration of the two endpoints gives

$$\begin{aligned} Z_\lambda &= \int_{-L}^{L} e^{-u_\lambda(y)} b_0(y)^2 \mathrm{d}y, \\ Z_\lambda\langle\phi_2^\ell\rangle &= \int_{-L}^{L} y^\ell e^{-u_\lambda(y)} b_0(y)^2 \mathrm{d}y, \quad \ell = 2, 4, \\ Z_\lambda\langle\phi_1\phi_3\rangle &= \int_{-L}^{L} e^{-u_\lambda(y)} b_1(y)^2 \mathrm{d}y. \end{aligned} \tag{S.49}$$

Nested adaptive vector quadrature evaluates the inner and outer integrals, separately from the tensor grid. The free-energy difference is $-\log(Z_\lambda/Z_0)$ with Z_0 evaluated on the same cutoff domain. The connected fourth cumulant is $\kappa_4 = \langle\phi_2^4\rangle - 3\langle\phi_2^2\rangle^2$; the parity-symmetric original law has zero mean. For the constrained law, the code centers the moments before computing a fourth cumulant, since its mean need not vanish.

The constrained three-index law uses unnormalized action masses $u_{ijk} = r_{0,ijk}e^{-V_{ijk}}$ and endpoint targets $r_i \propto w_i e^{-(x_i+0.50)^2/(2\,0.70^2)}$ and $s_k \propto w_k e^{-(x_k-0.45)^2/(2\,0.65^2)}$. Starting with zero log scalings, alternating marginal updates give

$$q_{ijk} = e^{\alpha_i+\beta_k}u_{ijk}, \qquad \alpha_i \leftarrow \alpha_i + \log r_i - \log\sum_{jk} q_{ijk}, \qquad \beta_k \leftarrow \beta_k + \log s_k - \log\sum_{ij} q_{ijk}. \tag{S.50}$$

Each update uses the current law, and all sums use log-sum-exp. The conditional distribution of y at fixed (x, z) is unchanged; hence the path and endpoint KL values agree. Dividing q by the endpoint factors and normalizing recovers p_h, checking Eq. (S.13). An intermediate iterate that violates one marginal is not a feasible primal solution, so its objective minus a dual value is not labeled a certified duality gap. The reported primal–dual difference is checked only at converged marginals.

The two methods agree closely for the stated finite problem, with field-quadrature, optimizer, marginal, and cutoff comparisons reported separately in the data files. Roundoff-size differences are empirical checks, not rigorous integration intervals. The calculation does not demonstrate improved sampling cost, large-volume scaling, a fermionic entropy solve, or a Lorentzian scattering amplitude.

Matched-cutoff perturbative comparison

This comparison uses exactly the action, couplings, and field domain in Eq. (S.46). Write $V_\lambda = \lambda U$, where $U = (x^4 + y^4 + z^4)/24$. The subscript zero below denotes the normalized quadratic reference on $[-L, L]^3$, not an unbounded Gaussian law. For a polynomial observable O the numerator and denominator have the local expansions

$$\begin{aligned} N_O(\lambda) &= \langle O\rangle_0 - \lambda\langle OU\rangle_0 + \tfrac{\lambda^2}{2}\langle OU^2\rangle_0 + O(\lambda^3), \\ D(\lambda) &= 1 - \lambda u_1 + \tfrac{\lambda^2}{2}u_2 + O(\lambda^3), \qquad u_j = \langle U^j\rangle_0, \\ D(\lambda)^{-1} &= 1 + \lambda u_1 + \lambda^2(u_1^2 - u_2/2) + O(\lambda^3). \end{aligned} \tag{S.51}$$

The bounded field domain justifies differentiation at zero coupling; no assertion about the usefulness of a finite truncation at a given nonzero coupling follows from that local expansion. Multiplying the series, rather than taking an unexpanded quotient of truncated polynomials, gives

$$\begin{aligned} \langle O\rangle_\lambda &= A_0(O) + \lambda A_1(O) + \lambda^2 A_2(O) + O(\lambda^3), \\ A_0(O) &= \langle O\rangle_0, \qquad A_1(O) = -\operatorname{Cov}_0(O, U), \\ A_2(O) &= \tfrac{1}{2}\operatorname{Cov}_0(O, U^2) - u_1 \operatorname{Cov}_0(O, U). \end{aligned} \tag{S.52}$$

Both normalization corrections are included in the code. First order retains $A_0 + \lambda A_1$; second order retains that expression plus $\lambda^2 A_2$. The free-energy expansion uses the same reference moments:

$$\Delta F(\lambda) = -\log\langle e^{-\lambda U}\rangle_0 = \lambda u_1 - \frac{\lambda^2}{2}(u_2 - u_1^2) + O(\lambda^3). \tag{S.53}$$

Since $U \geq 0$ on the domain, the untruncated value is nonnegative for $\lambda \geq 0$. A negative truncated value does not contradict this bound; it shows that the truncation is inaccurate there.

For the connected fourth cumulant, parity gives zero mean at every coupling. Let $v_j = A_j(y^2)$ and $m_j = A_j(y^4)$. Consistent re-expansion gives

$$\kappa_4^{(2)}(\lambda) = m_0 - 3v_0^2 + \lambda(m_1 - 6v_0v_1) + \lambda^2\{m_2 - 3(v_1^2 + 2v_0v_2)\}. \tag{S.54}$$

It is not $\langle y^4\rangle^{(2)} - 3(\langle y^2\rangle^{(2)})^2$ without truncating again, which would retain spurious third- and fourth-order terms. The finite cutoff makes $m_0 - 3v_0^2$ small but not identically zero; it is retained rather than set to zero by an unbounded-Gaussian identity.

The coefficients are computed with tensor Gauss–Legendre quadrature at $n = 80$ and checked again at $n = 96$. The supplied execution has maximum absolute change below 10^{-12}. The central variance polynomial has coefficients, rounded here,

$$(v_0, v_1, v_2) = (0.548387096771, -0.117191927112,\ 0.073149963081). \tag{S.55}$$

For the cumulant they are approximately $(-1.15 \times 10^{-11}, -0.0956372397970, 0.135092562747)$. Full-precision coefficients, all four compared observables, relative discrepancies, and adaptive-reference estimates are supplied in `data/nonperturbative/comparison.json` and its CSV companion. The full values are obtained by 100 mirror steps with step 1/4 at $n = 80$; the reference uses independently implemented nested adaptive quadrature. Both retain the full quartic interaction. First- and second-order results use only the zero-coupling coefficients above.

Figure S1 verifies the numerical minimization, whereas Fig. S2 shows the observable-level distinction from the particular perturbative truncations tested. At $\lambda = 4$, the second-order variance is 1.25001880 rather than 0.37529866, and its free-energy value is -0.16815104 rather than 0.34379167. These statements do not demonstrate a failure of perturbation theory at all orders, an advantage over resummation, or a computational advantage over direct integration or conventional sampling. The small non-Gaussian model supplies a nonperturbative finite-regulator check of the full entropy functional; the scalar/Yukawa and three-dimensional packet calculations remain separate scattering-response tests.

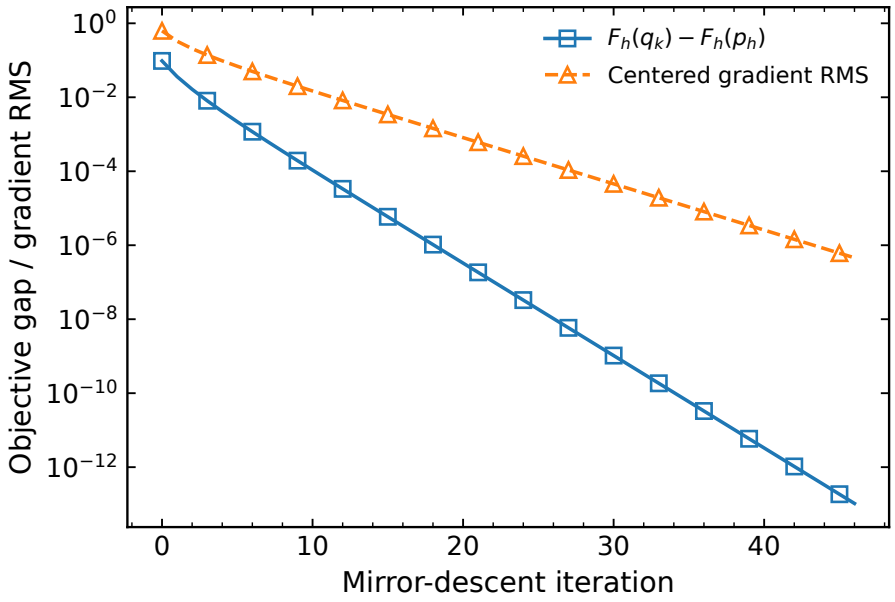


Figure S1: Optimization convergence retained from the preceding revision, $\lambda = 2$, $n = 48$. Hollow squares show $F_h(q_k) - F_h(p_h)$; hollow triangles show the centered gradient RMS. Objective gaps at the roundoff scale are omitted from the logarithmic display, with all iterations retained in the CSV. This figure tests optimization error at fixed quadrature, separately from field-grid refinement and the perturbative comparison.

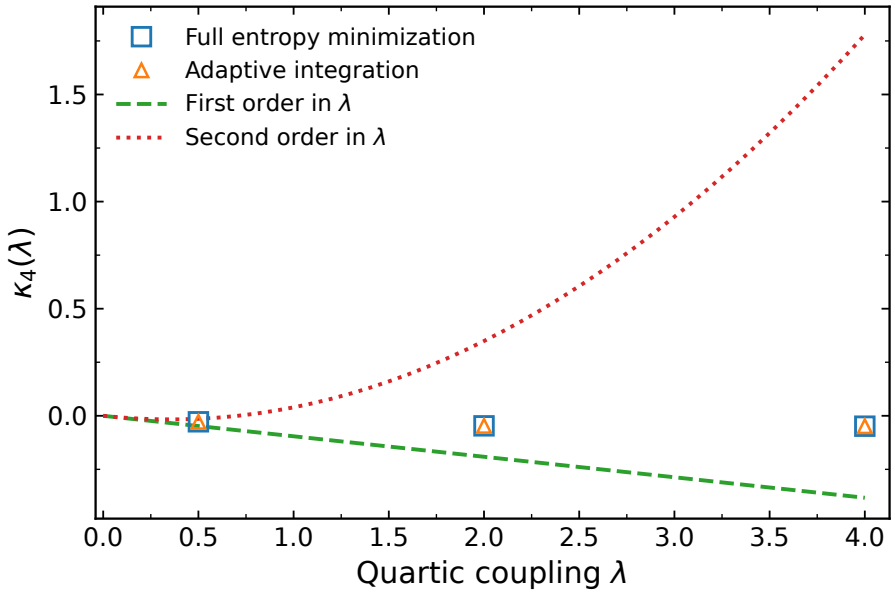


Figure S2: Connected fourth cumulant in the matched-cutoff three-site model. Hollow squares and triangles are the full entropy solve and independent adaptive integral. Lines are the consistently re-expanded first- and second-order expressions; no higher-order or resummed method is tested. At $\lambda = 2$ the full value is -0.04691680, whereas the second-order truncation is $+0.34909577$. The comparison concerns a finite Euclidean observable, not an amputated Lorentzian scattering amplitude.

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